Transformations, Domain & Range, and Growth/Decay of Exponential Functions

Exponential Functions Practice Worksheet

Name: _____ Date: _____ Class: _____

Section 1: Transformations of Exponential Functions

Directions: Circle the letter of the best answer.

Worked Example

Describe how $g(x) = 2^{x+1} - 5$ is transformed from the parent function $f(x) = 2^x$, and give the equation of its horizontal asymptote.

Solution: Compare $g(x) = 2^{x+1} - 5$ to the form $f(x - h) + k$ with $h = -1$ and $k = -5$.

- The $+1$ inside the exponent shifts the graph **left 1 unit**.
- The -5 outside shifts the graph **down 5 units**.
- Since $f(x) = 2^x$ has horizontal asymptote $y = 0$, shifting down 5 gives asymptote $\boxed{y = -5}$.

1. Compared to the parent function $f(x) = 2^x$, how is $g(x) = 2^x - 3$ transformed?

A. Shift left 3 units
B. Shift right 3 units
C. Shift up 3 units
D. Shift down 3 units

2. Compared to $f(x) = 3^x$, how is $g(x) = 3^{x+4}$ transformed?

A. Shift left 4 units
B. Shift right 4 units
C. Shift up 4 units
D. Shift down 4 units

3. Compared to $f(x) = 2^x$, how is $g(x) = -2^x$ transformed?

A. Reflection over the x-axis
B. Reflection over the y-axis
C. Vertical stretch
D. Horizontal shift

4. Compared to $f(x) = 5^x$, how is $g(x) = 5^{-x}$ transformed?

A. Reflection over the x-axis
B. Reflection over the y-axis
C. Vertical compression
D. Horizontal stretch

5. Compared to $f(x) = 2^x$, how is $g(x) = 4 \cdot 2^x$ transformed?

A. Vertical stretch by a factor of 4
B. Horizontal stretch by a factor of 4
C. Vertical shift up 4
D. Horizontal shift left 4

6. Compared to $f(x) = 5^x$, how is $g(x) = \frac{1}{3} \cdot 5^x$ transformed?

A. Vertical stretch
B. Vertical compression
C. Horizontal compression
D. Reflection over the x-axis

7. The parent function $f(x) = 2^x$ is shifted right 5 units and up 2 units. Which equation represents this transformation?

A. $g(x) = 2^{x-5} + 2$
B. $g(x) = 2^{x+5} + 2$
C. $g(x) = 2^{x-5} - 2$
D. $g(x) = 2^{x+2} - 5$

8. The parent function $f(x) = 3^x$ is reflected over the x-axis and shifted down 1 unit. Which equation represents this transformation?

A. $g(x) = -3^x - 1$
B. $g(x) = -3^x + 1$
C. $g(x) = 3^{-x} - 1$
D. $g(x) = 3^x - 1$

9. What is the equation of the horizontal asymptote of $g(x) = 2^{x-1} + 3$?

A. $y = 1$
B. $y = -1$
C. $y = 3$
D. $y = -3$

10. Which function represents $f(x) = b^x$ shifted down 4 units, with no other changes?

A. $g(x) = b^x - 4$
B. $g(x) = b^{x-4}$
C. $g(x) = b^x + 4$
D. $g(x) = -b^x - 4$

Section 2: Domain and Range of Transformed Exponential Functions

Directions: Circle the letter of the best answer.

Worked Example

Find the domain and range of $g(x) = 3^{x-2} + 4$.
Solution: Exponential functions b^{x-h} are defined for every real input, and transformations (shifts, stretches, reflections) never restrict that. So the **domain is all real numbers**.
For the range, look at a and k in $a \cdot b^{x-h} + k$: here $a = 1 > 0$ and $k = 4$. Since $b^{x-2} > 0$ for all x, we get $g(x) = 3^{x-2} + 4 > 4$. So the **range is** $y > 4$.

1. What is the domain of $f(x) = 2^x$?

A. $x > 0$
B. All real numbers
C. $x \neq 0$
D. $x \geq 0$

2. What is the range of $f(x) = 2^x$?

A. $y > 0$
B. All real numbers
C. $y \geq 0$
D. $y < 0$

3. What is the range of $g(x) = 2^x + 5$?

A. $y > 5$
B. $y > 0$
C. $y \geq 5$
D. All real numbers

4. What is the range of $g(x) = 2^x - 3$?

A. $y > -3$
B. $y < -3$
C. $y \geq -3$
D. All real numbers

5. What is the range of $g(x) = -2^x$?

A. $y > 0$
B. $y < 0$
C. $y \geq 0$
D. All real numbers

6. What is the range of $g(x) = -3^x + 4$?

A. $y < 4$
B. $y > 4$
C. $y \leq 4$
D. $y \geq 4$

7. What is the domain of $g(x) = 2^{x-3}$?

A. $x > 3$
B. All real numbers
C. $x \geq 3$
D. $x \neq 3$

8. What is the range of $g(x) = 5^{x+2} - 1$?

A. $y > -1$
B. $y < -1$
C. $y \geq -1$
D. All real numbers

9. For any function of the form $g(x) = a \cdot b^{x-h} + k$ (where $b > 0$, $b \neq 1$, $a \neq 0$), what is the domain?

A. All real numbers
B. $x > h$
C. $x \neq h$
D. It depends on the value of k

10. What is the range of $g(x) = -\left(\frac{1}{2}\right)^x - 6$?

A. $y < -6$
B. $y > -6$
C. $y \leq -6$
D. $y \geq -6$

Section 3: Exponential Growth and Decay

*Directions: Circle the letter of the best answer. Recall: for $y = a(1 + r)^t$, the quantity $(1 + r)$ is the **growth factor** and r is the **rate of growth**; for $y = a(1 - r)^t$, the quantity $(1 - r)$ is the **decay factor** and r is the **rate of decay**.*

Worked Example

A town's population is modeled by $P(t) = 12{,}000(1.045)^t$, where t is measured in years. Identify the growth factor and the rate of growth.
Solution: Match $P(t) = 12{,}000(1.045)^t$ to the form $a(1 + r)^t$. The base is 1.045, so the **growth factor is** 1.045. Since $1 + r = 1.045$, we get $r = 0.045$, so the **rate of growth is** 4.5%.

1. The function $y = 500(1.08)^t$ models growth. What is the growth factor?

A. 1.08 B. 0.08
C. 8 D. 500

2. The function $y = 500(1.08)^t$ models growth. What is the rate of growth?

A. 108% B. 8%
C. 1.08% D. 0.8%

3. The function $y = 200(0.95)^t$ models decay. What is the decay factor?

A. 0.95 B. 0.05
C. 5% D. 95

4. The function $y = 200(0.95)^t$ models decay. What is the rate of decay?

A. 95% B. 5%
C. 0.95% D. 0.05%

5. A population grows at a rate of 3% per year. What is the growth factor?

A. 1.03 B. 0.03
C. 3 D. 0.97

6. A car depreciates (decreases in value) by 12% per year. What is the decay factor?

A. 1.12 B. 0.12
C. 0.88 D. 12

7. The function $y = 1000(1.045)^t$ models growth. What is the rate of growth?

A. 1.045% B. 4.5%
C. 45% D. 104.5%

8. The function $y = 800(0.82)^t$ models decay. What is the rate of decay?

A. 82%
B. 8.2%
C. 18%
D. 0.18%

9. If the growth factor of an exponential model is 1.25, what is the rate of growth?

A. 125%
B. 1.25%
C. 25%
D. 2.5%

10. If the decay factor of an exponential model is 0.6, what is the rate of decay?

A. 60%
B. 40%
C. 6%
D. 4%

Section 4: Solving Exponential Equations

Directions: Solve each equation for x. Show your work in the space provided.

Worked Example

Solve for x: $4^{x+1} = 32$.
Solution: Rewrite both sides with base 2: $4 = 2^2$ and $32 = 2^5$.

$$4^{x+1} = \left(2^2\right)^{x+1} = 2^{2x+2}, \qquad 32 = 2^5$$

Since the bases match, set the exponents equal:

$$2x + 2 = 5 \implies 2x = 3 \implies x = \tfrac{3}{2}$$

1. Solve for x: $2^x = 32$

2. Solve for x: $3^{x+1} = 81$

3. Solve for x: $5^{2x} = 125$

4. Solve for x: $4^x = 64$

5. Solve for x: $2^{x-3} = 16$

6. Solve for x: $9^x = 27$

7. Solve for x: $7^{3x} = 7^{x+8}$

8. Solve for x: $2^{x+2} = 2^{3x-4}$

9. Solve for x: $8^x = 16$

10. Solve for x: $3^{2x-1} = 27^{x-2}$

Answer Key

Section 1: Transformations of Exponential Functions

1. **D.** Subtracting 3 outside the exponent shifts the graph down 3 units.
2. **A.** Adding 4 inside the exponent $(x + 4)$ shifts the graph left 4 units.
3. **A.** The negative sign in front of 2^x flips the graph over the x-axis.
4. **B.** Replacing x with $-x$ flips the graph over the y-axis.
5. **A.** Multiplying the whole function by 4 stretches it vertically by a factor of 4.
6. **B.** Multiplying by $\frac{1}{3}$ (a factor between 0 and 1) compresses the graph vertically.
7. **A.** A shift right 5 replaces x with $x - 5$; a shift up 2 adds 2: $g(x) = 2^{x-5} + 2$.
8. **A.** A reflection over the x-axis negates the function; a shift down 1 subtracts 1: $g(x) = -3^x - 1$.
9. **C.** The parent asymptote $y = 0$ shifts up 3 units (from the $+3$), giving $y = 3$. (The $x - 1$ only shifts horizontally and does not affect the asymptote.)
10. **A.** A vertical shift down 4 only changes the outside constant: $g(x) = b^x - 4$.

Section 2: Domain and Range of Transformed Exponential Functions

1. **B.** Exponential functions accept any real input, so the domain is all real numbers.
2. **A.** 2^x is always positive and gets arbitrarily close to (but never equals) 0, so range is $y > 0$.
3. **A.** Shifting 2^x $(y > 0)$ up 5 gives $y > 5$.
4. **A.** Shifting 2^x $(y > 0)$ down 3 gives $y > -3$.
5. **B.** Reflecting 2^x $(y > 0)$ over the x-axis flips the inequality: $y < 0$.
6. **A.** Reflecting 3^x gives $y < 0$; shifting up 4 gives $y < 4$.
7. **B.** Horizontal shifts never restrict the domain; it remains all real numbers.
8. **A.** $5^{x+2} > 0$ for all x; subtracting 1 gives $y > -1$.
9. **A.** No transformation (shift, stretch, or reflection) of b^x restricts which x-values are allowed.
10. **A.** $\left(\frac{1}{2}\right)^x > 0$, so $-\left(\frac{1}{2}\right)^x < 0$; subtracting 6 gives $y < -6$.

Section 3: Exponential Growth and Decay

1. **A.** In $y = a(1 + r)^t$, the base itself is the growth factor: 1.08.
2. **B.** Since $1 + r = 1.08$, $r = 0.08 = 8\%$.
3. **A.** In $y = a(1 - r)^t$, the base is the decay factor: 0.95.

4. **B.** Since $1 - r = 0.95$, $r = 0.05 = 5\%$.
5. **A.** $r = 0.03 \Rightarrow$ growth factor $= 1 + r = 1.03$.
6. **C.** $r = 0.12 \Rightarrow$ decay factor $= 1 - r = 1 - 0.12 = 0.88$.
7. **B.** Growth factor $1.045 \Rightarrow r = 0.045 = 4.5\%$.
8. **C.** Decay factor $0.82 \Rightarrow r = 1 - 0.82 = 0.18 = 18\%$.
9. **C.** Growth factor $1.25 = 1 + r \Rightarrow r = 0.25 = 25\%$.
10. **B.** Decay factor $0.6 = 1 - r \Rightarrow r = 0.4 = 40\%$.

Section 4: Solving Exponential Equations

1. $2^x = 32 = 2^5 \Rightarrow x = 5$
2. $3^{x+1} = 81 = 3^4 \Rightarrow x + 1 = 4 \Rightarrow x = 3$
3. $5^{2x} = 125 = 5^3 \Rightarrow 2x = 3 \Rightarrow x = \frac{3}{2}$
4. $4^x = 64 = 4^3 \Rightarrow x = 3$
5. $2^{x-3} = 16 = 2^4 \Rightarrow x - 3 = 4 \Rightarrow x = 7$
6. $9^x = (3^2)^x = 3^{2x}$ and $27 = 3^3 \Rightarrow 2x = 3 \Rightarrow x = \frac{3}{2}$
7. $7^{3x} = 7^{x+8} \Rightarrow 3x = x + 8 \Rightarrow 2x = 8 \Rightarrow x = 4$
8. $2^{x+2} = 2^{3x-4} \Rightarrow x + 2 = 3x - 4 \Rightarrow 6 = 2x \Rightarrow x = 3$
9. $8^x = (2^3)^x = 2^{3x}$ and $16 = 2^4 \Rightarrow 3x = 4 \Rightarrow x = \frac{4}{3}$
10. $3^{2x-1} = 27^{x-2} = (3^3)^{x-2} = 3^{3x-6} \Rightarrow 2x - 1 = 3x - 6 \Rightarrow x = 5$